

- Candidate features evaluated: 32
- Features removed by correlation pruning: 7
- Features kept after IC screening: 16
- Correlation threshold: 0.8
- IC threshold: 0.03
- IC horizons: future 1 week = 5 trading days, future 1 month = 21 trading days

- Trailing dividend features were converted to trailing dividend yield using current price, which is the standard BSM-style input.
- Historical volatility now includes 5-day, 20-day, and 60-day windows.
- Volatility change features were added using both the one-day difference and one-day percentage change of 20-day volatility.
- Redundant features with absolute Pearson correlation above 0.8 were removed before IC screening.
- Only features with absolute IC above 0.03 for at least one forecast horizon were retained.

- dgs10_momentum_5d
- jpm_vol_20d_change_rate_1d
- vix_jpm_corr_20d
- jpm_return_1d
- jpm_return_5d
- jpm_vol_5d
- jpm_vol_60d
- news_article_count
- jpm_price_to_ma_20d
- news_7d_sentiment_mean
- drawdown_20d
- dgs10
- news_7d_article_count
- jpm_dividend_growth_yoy
- jpm_dividend_ttm
- jpm dividend yield ttm

Open	High	Low	Close	Adj Close	Volume	dgs10	vix	jpm_return_1d	jpm_return_5d
jpm_vol_20d	jpm_vol_5d	jpm_vol_60d	jpm_vol_20d_change_1d	jpm_vol_20d_change_rate_1d					
jpm_ma_20d	jpm_price_to_ma_20d	dgs10_change_1d	dgs10_momentum_5d	vix_change_1d					
vix_jpm_corr_20d	rolling_high_20d	drawdown_20d	jpm_dividend_ttm	jpm_dividend_growth_yoy					
news_article_count	news_sentiment_mean	news_sentiment_std	news_7d_article_count						
news_7d_sentiment_mean	news_7d_sentiment_trend	jpm_dividend_yield_ttm							
Open		1.000	0.984	0.980	0.971	0.962	-0.398	0.477	-0.310
0.002	0.093	0.123	0.054	0.103			0.026		
0.063	0.952	0.194	0.027	0.071	0.014				
0.119	0.922	0.200	0.635	-0.517					
0.284	0.135	0.142	0.444	0.327					
-0.006	-0.851								

Week 2 Feature Engineering Optimization Report

High	0.984	1.000	0.983	0.986	0.972	-0.398	0.474	-0.305
0.003	0.143	0.129	0.039	0.115		-0.041		
-0.004	0.947	0.236	0.030		0.072	0.012		
0.115	0.921	0.223	0.636		-0.515			
0.284	0.140	0.143		0.442		0.328		
-0.003	-0.860							
Low	0.980	0.983	1.000	0.990	0.967	-0.407	0.480	-0.321
0.020	0.111	0.113	0.059	0.091		0.033		
0.069	0.951	0.212	0.032		0.074	0.004		
0.118	0.919	0.225	0.631		-0.523			
0.283	0.138	0.148		0.437		0.327		
-0.003	-0.862							
Close	0.971	0.986	0.990	1.000	0.977	-0.405	0.480	-0.317
0.040	0.148	0.136	0.070	0.112		0.005		
0.005	0.951	0.242	0.030		0.075	0.003		
0.120	0.924	0.234	0.636		-0.517			
0.282	0.140	0.150		0.442		0.331		
-0.003	-0.867							
Adj Close	0.962	0.972	0.967	0.977	1.000	-0.409	0.529	-0.275
0.138	0.192	0.213	0.144	0.193		0.069		
0.010	0.962	0.283	0.029		0.075	0.002		
0.145	0.950	0.215	0.707		-0.523			
0.299	0.146	0.170		0.476		0.343		
-0.004	-0.834							
Volume	-0.398	-0.398	-0.407	-0.405	-0.409	1.000	-0.376	0.422
-0.035	-0.104	-0.037	0.004	-0.006		0.055		
0.143	-0.391	-0.147	-0.029		-0.084	0.062		
0.038	-0.367	-0.245	-0.238		0.352			
-0.041	-0.055	-0.024		-0.089		-0.128		
-0.005	0.430							
dgs10	0.477	0.474	0.480	0.480	0.529	-0.376	1.000	-0.365
0.030	0.049	0.137	0.088	0.089		0.048		
0.038	0.538	0.057	0.038		0.085	-0.002		
-0.046	0.530	0.076	0.599		-0.496			
0.068	0.035	0.061		0.112		0.081		
0.001	-0.462							
vix	-0.310	-0.305	-0.321	-0.317	-0.275	0.422	-0.365	1.000
-0.051	-0.122	-0.013	-0.004	0.049		-0.020		
0.015	-0.272	-0.152	-0.019		-0.008	0.171		
-0.102	-0.247	-0.280	0.024		0.360			
0.128	-0.001	0.116		0.188		0.004		
-0.008	0.599							
jpm_return_1d	0.002	0.003	0.020	0.040	0.138	-0.035	0.030	-0.051
1.000	0.446	0.171	0.251	0.149		0.530		
-0.072	0.028	0.378	0.044		0.016	-0.083		
0.027	0.059	0.232	0.037		-0.005			
0.012	0.045	0.008		0.032		0.024		
0.019	-0.123							

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jpm_return_5d	0.093	0.143	0.111	0.148	0.192	-0.104	0.049	-0.122
0.446	1.000	0.220	0.280	0.224	0.025			
-0.144	0.014	0.673	0.048	0.139	-0.023			
0.063	0.076	0.395	0.049	-0.037				
0.016	0.082	0.002	0.042	0.104				
0.050	-0.180							
jpm_vol_20d	0.123	0.129	0.113	0.136	0.213	-0.037	0.137	-0.013
0.171	0.220	1.000	0.736	0.975	0.159			
0.061	0.205	0.075	0.012	0.026	0.001			
0.307	0.380	-0.446	0.241	0.131				
0.050	0.020	0.014	0.108	0.051				
-0.001	-0.000							
jpm_vol_5d	0.054	0.039	0.059	0.070	0.144	0.004	0.088	-0.004
0.251	0.280	0.736	1.000	0.654	0.362			
0.140	0.157	-0.007	0.007	0.004	-0.015			
0.245	0.278	-0.376	0.175	0.118				
0.063	0.018	0.030	0.106	0.039				
-0.012	0.021							
jpm_vol_60d	0.103	0.115	0.091	0.112	0.193	-0.006	0.089	0.049
0.149	0.224	0.975	0.654	1.000	0.101			
0.039	0.158	0.154	0.019	0.045	0.002			
0.322	0.335	-0.381	0.225	0.176				
0.032	0.015	0.001	0.086	0.048				
-0.004	0.040							
jpm_vol_20d_change_1d	0.026	-0.041	0.033	0.005	0.069	0.055	0.048	-0.020
0.530	0.025	0.159	0.362	0.101	1.000			
0.589	0.097	-0.068	-0.014	-0.023	-0.012			
0.080	0.099	-0.091	0.058	0.020				
0.002	0.016	0.033	0.043	0.015				
-0.004	-0.033							
jpm_vol_20d_change_rate_1d	0.063	-0.004	0.069	0.005	0.010	0.143	0.038	0.015
-0.072	-0.144	0.061	0.140	0.039	0.589			
1.000	0.075	-0.178	0.037	-0.009	-0.005			
0.083	0.072	-0.180	0.046	0.019				
0.019	0.026	0.057	0.030	0.015				
0.027	0.017							
jpm_ma_20d	0.952	0.947	0.951	0.951	0.962	-0.391	0.538	-0.272
0.028	0.014	0.205	0.157	0.158	0.097			
0.075	1.000	0.020	0.015	0.041	0.003			
0.144	0.979	0.016	0.729	-0.531				
0.301	0.132	0.170	0.477	0.310				
-0.006	-0.780							
jpm_price_to_ma_20d	0.194	0.236	0.212	0.242	0.283	-0.147	0.057	-0.152
0.378	0.673	0.075	-0.007	0.154	-0.068			
-0.178	0.020	1.000	0.081	0.176	-0.007			
0.046	0.052	0.764	0.034	-0.101				
0.032	0.082	0.028	0.063	0.178				
0.019	-0.311							

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dgs10_change_1d	0.027	0.030	0.032	0.030	0.029	-0.029	0.038	-0.019
0.044	0.048	0.012	0.007	0.019		-0.014		
0.037	0.015	0.081	1.000		0.441	-0.069		
0.016	0.019	0.056	0.048		-0.052			
0.004	0.019	0.002		0.026		-0.001		
0.015	-0.014							
dgs10_momentum_5d	0.071	0.072	0.074	0.075	0.075	-0.084	0.085	-0.008
0.016	0.139	0.026	0.004	0.045		-0.023		
-0.009	0.041	0.176	0.441	1.000	0.012			
0.032	0.048	0.128	0.113		-0.123			
0.017	0.030	0.029		0.052	0.041			
0.037	-0.036							
vix_change_1d	0.014	0.012	0.004	0.003	0.002	0.062	-0.002	0.171
-0.083	-0.023	0.001	-0.015	0.002	-0.012			
-0.005	0.003	-0.007	-0.069	0.012	1.000			
-0.009	0.003	-0.009	0.007	0.009				
0.007	0.001	-0.006		-0.006	0.004			
-0.003	-0.011							
vix_jpm_corr_20d	0.119	0.115	0.118	0.120	0.145	0.038	-0.046	-0.102
0.027	0.063	0.307	0.245	0.322	0.080			
0.083	0.144	0.046	0.016	0.032	-0.009			
1.000	0.195	-0.141	0.149	0.151				
0.019	0.036	0.003		0.011	0.108			
-0.007	0.044							
rolling_high_20d	0.922	0.921	0.919	0.924	0.950	-0.367	0.530	-0.247
0.059	0.076	0.380	0.278	0.335	0.099			
0.072	0.979	0.052	0.019	0.048	0.003			
0.195	1.000	-0.082	0.743	-0.472				
0.302	0.134	0.169		0.480	0.314			
-0.004	-0.730							
drawdown_20d	0.200	0.223	0.225	0.234	0.215	-0.245	0.076	-0.280
0.232	0.395	-0.446	-0.376	-0.381	-0.091			
-0.180	0.016	0.764	0.056	0.128	-0.009			
-0.141	-0.082	1.000	-0.074	-0.272				
-0.002	0.058	0.010		0.000	0.117			
0.016	-0.411							
jpm_dividend_ttm	0.635	0.636	0.631	0.636	0.707	-0.238	0.599	0.024
0.037	0.049	0.241	0.175	0.225	0.058			
0.046	0.729	0.034	0.048	0.113	0.007			
0.149	0.743	-0.074	1.000	-0.540				
0.298	0.115	0.213		0.490	0.277			
-0.006	-0.268							
jpm_dividend_growth_yoy	-0.517	-0.515	-0.523	-0.517	-0.523	0.352	-0.496	0.360
-0.005	-0.037	0.131	0.118	0.176	0.020			
0.019	-0.531	-0.101	-0.052	-0.123	0.009			
0.151	-0.472	-0.272	-0.540	1.000				
-0.197	-0.058	-0.115		-0.342	-0.133			
0.001	0.571							

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news_article_count	0.284	0.284	0.283	0.282	0.299	-0.041	0.068	0.128
0.012	0.016	0.050	0.063	0.032	0.002			
0.019	0.301	0.032	0.004	0.017	0.007			
0.019	0.302	-0.002	0.298	-0.197				
1.000	0.265	0.557	0.635	0.319				
0.108	-0.109							
news_sentiment_mean	0.135	0.140	0.138	0.140	0.146	-0.055	0.035	-0.001
0.045	0.082	0.020	0.018	0.015	0.016			
0.026	0.132	0.082	0.019	0.030	0.001			
0.036	0.134	0.058	0.115	-0.058				
0.265	1.000	0.086	0.218	0.441				
0.692	-0.085							
news_sentiment_std	0.142	0.143	0.148	0.150	0.170	-0.024	0.061	0.116
0.008	0.002	0.014	0.030	0.001	0.033			
0.057	0.170	0.028	0.002	0.029	-0.006			
0.003	0.169	0.010	0.213	-0.115				
0.557	0.086	1.000	0.397	0.181				
0.019	-0.001							
news_7d_article_count	0.444	0.442	0.437	0.442	0.476	-0.089	0.112	0.188
0.032	0.042	0.108	0.106	0.086	0.043			
0.030	0.477	0.063	0.026	0.052	-0.006			
0.011	0.480	0.000	0.490	-0.342				
0.635	0.218	0.397	1.000	0.508				
0.021	-0.191							
news_7d_sentiment_mean	0.327	0.328	0.327	0.331	0.343	-0.128	0.081	0.004
0.024	0.104	0.051	0.039	0.048	0.015			
0.015	0.310	0.178	-0.001	0.041	0.004			
0.108	0.314	0.117	0.277	-0.133				
0.319	0.441	0.181	0.508	1.000				
0.220	-0.204							
news_7d_sentiment_trend	-0.006	-0.003	-0.003	-0.003	-0.004	-0.005	0.001	-0.008
0.019	0.050	-0.001	-0.012	-0.004	-0.004			
0.027	-0.006	0.019	0.015	0.037	-0.003			
-0.007	-0.004	0.016	-0.006	0.001				
0.108	0.692	0.019	0.021	0.220				
1.000	0.000							
jpm_dividend_yield_ttm	-0.851	-0.860	-0.862	-0.867	-0.834	0.430	-0.462	0.599
-0.123	-0.180	-0.000	0.021	0.040	-0.033			
0.017	-0.780	-0.311	-0.014	-0.036	-0.011			
0.044	-0.730	-0.411	-0.268	0.571				
-0.109	-0.085	-0.001	-0.191	-0.204				
0.000	1.000							

Feature Score Table

feature	mean_abs_corr	ic_1w	ic_1m	max_abs_ic	corr_pruned	selected	drop_reason
drawdown_20d	0.192119	-0.364872	-0.486800	0.486800	False	True	
kept							
jpm_vol_20d	0.170726	0.191156	0.485115	0.485115	True	False	

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correlation>=0.8

jpm_vol_60d 0.159453 0.195044 0.475982 0.475982 False True

kept

jpm_price_to_ma_20d 0.162928 -0.272767 -0.456884 0.456884 False True

kept

jpm_return_5d 0.138510 -0.119657 -0.283053 0.283053 False True

kept

jpm_vol_20d_change_rate_1d 0.073739 0.262171 -0.005954 0.262171 False

True kept

jpm_dividend_yield_ttm 0.320867 0.152629 0.242706 0.242706 False True

kept

jpm_vol_5d 0.146920 0.117503 0.233906 0.233906 False True

kept

jpm_return_1d 0.100421 -0.202804 -0.188651 0.202804 False True

kept

Close 0.373820 -0.077916 -0.141411 0.141411 True False correlation>=0.8

Low 0.371355 -0.039111 -0.141051 0.141051 True False correlation>=0.8

High 0.370487 -0.053806 -0.132515 0.132515 True False correlation>=0.8

Adj Close 0.393918 -0.081853 -0.117715 0.117715 True False

correlation>=0.8

vix_jpm_corr_20d 0.098101 0.049177 0.117612 0.117612 False True

kept

Open 0.367358 -0.040673 -0.115618 0.115618 True False correlation>=0.8

jpm_dividend_growth_yoy 0.257293 -0.038366 -0.094466 0.094466 False True

kept

dgs10_momentum_5d 0.069708 0.000039 -0.072232 0.072232 False True

kept

news_7d_article_count 0.225636 0.022833 0.068919 0.068919 False True

kept

rolling_high_20d 0.380796 0.063334 0.059442 0.063334 True False

correlation>=0.8

jpm_dividend_ttm 0.291740 0.052335 0.060520 0.060520 False True

kept

news_article_count 0.162461 0.050648 0.038226 0.050648 False True

kept

news_7d_sentiment_mean 0.178082 -0.020879 -0.046304 0.046304 False True

kept

dgs10 0.224071 0.034736 0.044176 0.044176 False True kept

dgs10_change_1d 0.041773 0.020046 -0.028978 0.028978 False False IC

threshold

vix_change_1d 0.018534 -0.028418 -0.015596 0.028418 False False IC

threshold

jpm_vol_20d_change_1d 0.089347 0.012318 -0.027763 0.027763 False False

IC threshold

news_7d_sentiment_trend 0.042734 0.027513 -0.008841 0.027513 False False

IC threshold

vix 0.169954 -0.017727 0.026946 0.026946 False False IC threshold

news_sentiment_std 0.099496 0.021422 0.024189 0.024189 False False

IC threshold

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Volume 0.191068 -0.016303 0.022122 0.022122 False False IC threshold

IC-Passed Features

feature	ic_1w	ic_1m	max_abs_ic
drawdown_20d	-0.364872	-0.486800	0.486800
jpm_vol_60d	0.195044	0.475982	0.475982
jpm_price_to_ma_20d	-0.272767	-0.456884	0.456884
jpm_return_5d	-0.119657	-0.283053	0.283053
jpm_vol_20d_change_rate_1d	0.262171	-0.005954	0.262171
jpm_dividend_yield_ttm	0.152629	0.242706	0.242706
jpm_vol_5d	0.117503	0.233906	0.233906
jpm_return_1d	-0.202804	-0.188651	0.202804
vix_jpm_corr_20d	0.049177	0.117612	0.117612
jpm_dividend_growth_yoy	-0.038366	-0.094466	0.094466
dgs10_momentum_5d	0.000039	-0.072232	0.072232
news_7d_article_count	0.022833	0.068919	0.068919
jpm_dividend_ttm	0.052335	0.060520	0.060520
news_article_count	0.050648	0.038226	0.050648
news_7d_sentiment_mean	-0.020879	-0.046304	0.046304
dgs10	0.034736	0.044176	0.044176